Linear Algebra Problem Sheet-3

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September 2, 2026

Abstract

The third problem sheet for Graduate Mathematical Toolkit-1 2026. Please contact the TAs for doubts at sunain.ug2023@cmi.ac.in and Avishekh@cmi.ac.in. Sources are Axler's Linear algebra done right and Freud's Linear algebra A problem centered approach

1 Determinants

Problem 1.1. *Compute the $n \times n$ determinant with α_{ij} equal to*

1. *i if $i = j$ and 1 if $i \neq j$;*
2. *$\min(i, j)$;*
3. *ij;*
4. *$i + j$;*
5. *$i^2 + j^2$.*

Problem 1.2. *Let $n \geq 7$ and let α_{ij} be the least positive remainder of ij modulo n (so $\alpha_{ij} = n$, not 0, when $n \mid ij$). Verify that the determinant is 0.*

Problem 1.3. *The jth entry of the ith row of an $n \times n$ determinant D is 2^{ij}. What is the highest power of 2 dividing D?*

Problem 1.4. *Let $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_n$ satisfy $\alpha_i \beta_j \neq 1$ for all i, j. Compute the $n \times n$ determinant whose (i, j) entry is $\dfrac{1 - \alpha_i^n \beta_j^n}{1 - \alpha_i \beta_j}$.*

Problem 1.5. *Let $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_n$ be arbitrary. Compute the $n \times n$ determinant whose (i, j) entry is $(\alpha_i + \beta_j)^{n-1}$.*

Problem 1.6 ($*$)**.** *Let $p > 2$ be prime and $V_p = V(1, 2, \ldots, p)$ the Vandermonde matrix on the nodes $1, \ldots, p$. What is the remainder of $(\det V_p)^2$ modulo p?*

Problem 1.7. *Prove, without computing the determinant of these dates from history, that*

$$\begin{vmatrix} 1861 & 1492 & 1896 & 1984 \\ 1782 & 1929 & 1918 & 1624 \\ 1776 & 1850 & 1945 & 1756 \\ 1790 & 1914 & 1770 & 1787 \end{vmatrix} \neq 0.$$

Problem 1.8. *Let p be prime and let A be a $p \times p$ matrix over $\mathbb{F}_p$ with every diagonal entry equal to 1 and every entry below the diagonal equal to 0. Compute A^p.*

2 Permutations and inversions

Problem 2.1. *1. What is the largest possible change in the number of inversions caused by swapping two elements of a permutation, and when does it occur?*

(∗b) P chooses a permutation and S then swaps two of its elements. P maximizes the resulting change in the number of inversions, S minimizes it. What is the change under optimal play?

Problem 2.2. *P chooses a permutation and S must restore the identity by admissible steps; P maximizes the number of steps, S minimizes it. How many steps are needed under optimal play if a step is*

1. *swapping two elements of adjacent sizes (for example 6 and 7, wherever they stand);*

(∗b) swapping two arbitrary elements;

(∗c) swapping 1 with any other element?

Problem 2.3. *Let $f(n,k)$ be the number of permutations of $1,\dots,n$ with exactly k inversions.*

1. *Verify $f(n,k) = \sum_{i=k-n+1}^{k} f(n-1,i)$.*
2. *Prove $f(n,k) = f(n-1,k) + f(n,k-1) - f(n-1,k-n)$.*
3. *Compute $\sum_k f(n,k)$.*
4. *Find a simple formula for $\sum_k k \cdot f(n,k)$.*
5. *Show $\max_k f(n,k) \geq 2(n-2)!$ for $n > 2$.*
6. *For which k is $f(n,k)$ maximal, n fixed?*

3 More on Determinants

Problem 3.1 (∗). *Consider $n \times n$ matrices A with at most two nonzero entries in each row, and let $k(A)$ be the number of nonzero signed products of n factors in the Leibniz sum for $\det A$. What is the maximal possible value of $k(A)$?*

Problem 3.2 (∗). *1. P presents an $n \times n$ real matrix. In each step R replaces one entry by any real number, until the determinant is nonzero. P maximizes the number of steps, R minimizes it. How many steps are needed under optimal play?*

2. *Same game, except that R announces how many entries he wants to change, P then chooses those positions, and R replaces the entries there.*
3. *Discuss both games when R must instead reach determinant 0.*

Problem 3.3 (∗). *Does there exist, for every n, an $n \times n$ real matrix with determinant 0 such that modifying exactly one entry, arbitrarily, never makes the determinant nonzero?*

Problem 3.4. *Prove that $(A+B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$ and $(AB)^T = B^T A^T$ whenever one side is defined, and show the other side is then defined too. Formulate and verify the analogous statements for the adjoint.*

Problem 3.5. *Let A be a real invertible matrix such that all entries of A and of A^{-1} are nonnegative. Prove that every row and every column of A has exactly one nonzero entry.*

Problem 3.6. *For which integers n does there exist an $n \times n$ rational matrix with $A^3 = 2I$?*

4 Linear systems and rank

Problem 4.1. *Let $1 \leq m \leq n$. When does the $n \times n$ system*

$$x_1 + x_2 + \cdots + x_m = 1, \quad x_2 + x_3 + \cdots + x_{m+1} = 1, \quad \ldots, \quad x_n + x_1 + \cdots + x_{m-1} = 1$$

have a unique solution over $\mathbb{R}$?

Problem 4.2. *Let k, n be arbitrary and $1 \leq r \leq \min(n, k)$. Exhibit a $k \times n$ matrix of rank r in which no $r \times r$ minor is 0.*

Problem 4.3. *True or false, with proof or counterexample.*

1. *If $A\mathbf{x} = \mathbf{b}$ is solvable, then the columns of $[A \mid \mathbf{b}]$ are linearly dependent.*
2. *If the columns of $[A \mid \mathbf{b}]$ are linearly dependent, then $A\mathbf{x} = \mathbf{b}$ is solvable.*
3. *If the columns of A are linearly independent, then $A\mathbf{x} = \mathbf{b}$ is solvable.*
4. *If the rows of A are linearly independent, then $A\mathbf{x} = \mathbf{b}$ is solvable.*

Problem 4.4. *Is every real matrix a sum of matrices in which*

1. *every row is an arithmetic sequence;*
2. *every row or every column is an arithmetic sequence;*
3. *every row is a geometric sequence?*

Problem 4.5 ($*$). *P presents a system of k linear equations in n unknowns over $\mathbb{R}$. R must make it solvable by repeatedly replacing one coefficient or constant by another number. P maximizes the number of steps, R minimizes it. How many steps are needed under optimal play?*

Problem 4.6. *Let F be a finite field. Show that every function $\Phi : F \to F$ is a polynomial function, that is, $f(\tau) = \Phi(\tau)$ for all $\tau \in F$ for some $f \in F[x]$.*

Problem 4.7 ($*$). *Ali Baba wants to leave his treasure to the Forty Thieves, but only if at least 25 of them agree. The road to the treasure is guarded by a password which is a large integer. Ali Baba whispers something to each thief separately, so that any 25 of them can together recover the password, while any 24 of them have no information about it whatsoever. What was Interpol(ation)'s advice?*

5 Vector spaces and counting over $\mathbb{F}_p$

Problem 5.1.

1. *Does there exist a vector space with 243 elements over $\mathbb{F}_3$?*
2. *Does there exist one with 300 elements over $\mathbb{F}_3$?*

($**c$) Prove that the number of elements of a finite field is a prime power.

3. *Prove that the number of elements of a finite vector space is a prime power.*

Problem 5.2. *Assume a vector space has only finitely many bases, and at least one. Show it has finitely many elements.*

Problem 5.3 ($*$).

1. *How many bases does $\mathbb{F}_p^2$ have? And $\mathbb{F}_p^n$?*

2. *How many $n \times n$ matrices over $\mathbb{F}_p$ are invertible?*

3. *Prove $n! \mid (p^n-1)(p^n-p)\cdots(p^n-p^{n-1})$ for every prime p and every $n > 0$.*

Problem 5.4 ($**$). *How many $k \times n$ matrices over $\mathbb{F}_p$ have rank exactly r?*

Problem 5.5. 1. *Prove that if every entry of a matrix is 0 or 1, then its rank over $\mathbb{F}_2$ is at most its rank over $\mathbb{R}$.*

2. *Exhibit a 0–1 matrix whose rank over $\mathbb{F}_2$ is smaller than its rank over $\mathbb{R}$ by 1000.*

($**$c) Determine the smallest n for which an $n \times n$ matrix with the property in (b) exists.

6 Linear maps and Hom

Problem 6.1. *How many linear transformations $\mathcal{A}$ of $\mathbb{R}^2$ satisfy*

1. *$\mathcal{A}\binom{1}{3} = \binom{0}{0}$ and $\mathcal{A}\binom{2}{6} = \binom{1}{0}$;*

2. *$\mathcal{A}\binom{1}{3} = \binom{0}{0}$ and $\mathcal{A}\binom{2}{3} = \binom{1}{0}$;*

3. *$\mathcal{A}\binom{1}{3} = \binom{6}{5}$ and $\mathcal{A}\binom{2}{6} = \binom{12}{10}$?*

Problem 6.2. *Let $F = \mathbb{F}_p$. How many linear maps $\mathcal{A} : F^n \to F^k$ are there?*

Problem 6.3. *Let V_1, V_2 be finite-dimensional over F. Prove that there is a linear map $\mathcal{A} : V_1 \to V_2$ with $\operatorname{Ker}\mathcal{A} = \mathbf{0}$ or $\operatorname{Im}\mathcal{A} = V_2$.*

Problem 6.4. *A linear map $\mathcal{A} : V_1 \to V_2$ satisfies: the images of any four vectors are linearly dependent, and no six linearly independent vectors of V_1 are all mapped to zero. Prove $\dim V_1 \le 8$.*

Problem 6.5. *Let $\mathcal{A}, \mathcal{B} : V_1 \to V_2$ be linear with $\operatorname{Ker}\mathcal{A} \subseteq \operatorname{Ker}\mathcal{B}$ and $\operatorname{Im}\mathcal{A} \subseteq \operatorname{Im}\mathcal{B}$. Verify $\operatorname{Ker}\mathcal{A} = \operatorname{Ker}\mathcal{B}$ and $\operatorname{Im}\mathcal{A} = \operatorname{Im}\mathcal{B}$.*

Problem 6.6. *Let $\mathcal{A}, \mathcal{B} : V_1 \to V_2$ satisfy $V_1 = \operatorname{Ker}\mathcal{A} \oplus \operatorname{Ker}\mathcal{B}$ and $V_2 = \operatorname{Im}\mathcal{A} \oplus \operatorname{Im}\mathcal{B}$. Prove $V_1 \cong V_2$.*

Problem 6.7. *For $\mathcal{A}, \mathcal{B} \in \operatorname{Hom}(V_1, V_2)$ prove*

1. *$\operatorname{Ker}(\mathcal{A}+\mathcal{B}) \supseteq \operatorname{Ker}\mathcal{A} \cap \operatorname{Ker}\mathcal{B}$;*

2. *$\operatorname{Im}(\mathcal{A}+\mathcal{B}) \subseteq \langle \operatorname{Im}\mathcal{A}, \operatorname{Im}\mathcal{B} \rangle$;*

3. *$\operatorname{Ker}(\lambda\mathcal{A}) = \operatorname{Ker}\mathcal{A}$ for $\lambda \neq 0$;*

4. *$\operatorname{Im}(\lambda\mathcal{A}) = \operatorname{Im}\mathcal{A}$ for $\lambda \neq 0$.*

In (a) and (b) give examples of equality and of strict containment.

Problem 6.8. *Let $V_1 = V_2$ be the space of plane vectors. Describe $\mathcal{A}+\mathcal{B}$ when $\mathcal{A}$ and $\mathcal{B}$ are*

1. *the reflections across the x and y axes;*

2. *the perpendicular projections onto the x and y axes;*

3. *the rotations about the origin by $+60^\circ$ and -60°;*

4. *the rotations about the origin by Φ and $-\Phi$;*

5. *the identity and the rotation about the origin by $+90^\circ$.*

Problem 6.9. *Which of the following sets of linear maps form a subspace of* $\mathrm{Hom}(V_1, V_2)$*?*

1. *Those whose kernel is a fixed subspace $U_1 \subseteq V_1$.*

2. *Those whose image has dimension at most one.*

3. *Those for which every image is a scalar multiple of a fixed vector.*

4. *Those mapping a given subspace of V_1 into a given subspace $U_2 \subseteq V_2$.*

5. *Those mapping a given vector of V_1 to a given vector of V_2.*

Problem 6.10. *Let $V_1 = V_2 = V$. Which of the following form a subspace of* $\mathrm{Hom}\, V$*?*

1. *All isomorphisms.*

2. *All non-isomorphisms.*

3. *All transformations fixing a given vector.*

4. *All transformations whose image is contained in their kernel.*

Problem 6.11. *True or false, for $\mathcal{A}, \mathcal{B} \in \mathrm{Hom}(V_1, V_2)$?*

1. $\mathrm{Im}\,\mathcal{A} \cap \mathrm{Im}\,\mathcal{B} = \mathbf{0} \;\Rightarrow\; \mathrm{Ker}(\mathcal{A}+\mathcal{B}) = \mathrm{Ker}\,\mathcal{A} \cap \mathrm{Ker}\,\mathcal{B}$.

2. $\mathrm{Ker}(\mathcal{A}+\mathcal{B}) = \mathrm{Ker}\,\mathcal{A} \cap \mathrm{Ker}\,\mathcal{B} \;\Rightarrow\; \mathrm{Im}\,\mathcal{A} \cap \mathrm{Im}\,\mathcal{B} = \mathbf{0}$.

Problem 6.12. *Let $a_1, \ldots, a_n$ be a basis of V_1 and $b_1, \ldots, b_k$ a basis of V_2, and define $\mathcal{C}_{ij} \in \mathrm{Hom}(V_1, V_2)$ by $\mathcal{C}_{ij}a_r = b_i$ if $r = j$ and $\mathcal{C}_{ij}a_r = \mathbf{0}$ otherwise. Verify that the $\mathcal{C}_{ij}$ form a basis of* $\mathrm{Hom}(V_1, V_2)$*, and deduce* $\dim \mathrm{Hom}(V_1, V_2) = \dim V_1 \cdot \dim V_2$.

Problem 6.13. *True or false, for $\mathcal{A}_1, \ldots, \mathcal{A}_r \in \mathrm{Hom}(V_1, V_2)$?*

1. *If $\mathcal{A}_1, \ldots, \mathcal{A}_r$ are linearly dependent in* $\mathrm{Hom}(V_1, V_2)$*, then $\mathcal{A}_1 x, \ldots, \mathcal{A}_r x$ are linearly dependent in V_2 for every $x \in V_1$.*

2. *If $\mathcal{A}_1 x, \ldots, \mathcal{A}_r x$ are linearly dependent in V_2 for every $x \in V_1$, then $\mathcal{A}_1, \ldots, \mathcal{A}_r$ are linearly dependent in* $\mathrm{Hom}(V_1, V_2)$*.*

Problem 6.14. *Let $V_1 = \mathbb{C}^4$, $V_2 = \mathbb{C}^2$ and $\mathcal{A}_{ij}(\alpha_1, \alpha_2, \alpha_3, \alpha_4)^T = (\alpha_i, \alpha_j)^T$ for $1 \le i, j \le 4$.*

1. *Prove that any three distinct maps $\mathcal{A}_{ij}$ are linearly independent.*

2. *Exhibit four distinct linearly dependent maps $\mathcal{A}_{ij}$.*

(∗c) *What is the maximal number of linearly independent maps $\mathcal{A}_{ij}$?*

Problem 6.15. *Let $2 \le \dim V < \infty$ over an infinite field F and let $\mathcal{A} \in \mathrm{Hom}\, V$. Prove that infinitely many bases of V, no two scalar multiples of each other, induce the same matrix $[\mathcal{A}]$.*

7 Some more problems

Problem 7.1. *Let*

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}.$$

Compute the characteristic polynomial, the eigenvalues, and a basis of each eigenspace. Find an invertible P with $P^{-1}AP$ diagonal.

Problem 7.2. *For each of*

$$B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

decide whether it is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$, and justify.

Problem 7.3. *Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Show $A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, diagonalize A, and deduce a closed form for F_n.*

Problem 7.4. *Find all eigenvalues and eigenvectors of:*

1. *$T \in \mathcal{L}(\mathbb{R}^2)$, $T(x, y) = (-3y, x)$;*
2. *$T \in \mathcal{L}(\mathbb{F}^2)$, $T(w, z) = (z, w)$;*
3. *$T \in \mathcal{L}(\mathbb{F}^3)$, $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$.*

In (b) and (c), say how the answer depends on $\mathbb{F}$.

Problem 7.5. *Let T be a linear operator on a finite-dimensional space. Prove that eigenvectors belonging to distinct eigenvalues are linearly independent. Deduce that an operator on an n-dimensional space with n distinct eigenvalues is diagonalizable.*

Problem 7.6. *For $A \in M_n(\mathbb{C})$ with eigenvalues $\lambda_1, \dots, \lambda_n$ listed with algebraic multiplicity, prove $\operatorname{trace} A = \sum_i \lambda_i$ and $\det A = \prod_i \lambda_i$. Identify the remaining coefficients of the characteristic polynomial as sums of principal minors.*

Problem 7.7. *Prove that A and A^T have the same characteristic polynomial, and that A and $B^{-1}AB$ have the same eigenvalues for invertible B. Prove that AB and BA have the same nonzero eigenvalues with the same multiplicities, and that for $S, T \in \mathcal{L}(V)$ with $\dim V < \infty$ the operators ST and TS have the same eigenvalues. Give A, B with $AB \neq BA$.*

Problem 7.8. *Let $\mathbb{F}$ be algebraically closed and T an operator on $\mathbb{F}^n$. Prove T is nilpotent if and only if 0 is its only eigenvalue. Prove that the geometric multiplicity of an eigenvalue is at most its algebraic multiplicity, and give an example where the inequality is strict.*

Problem 7.9. *Show that the forward shift $T \in \mathcal{L}(\mathbb{F}^\infty)$, $T(z_1, z_2, \dots) = (0, z_1, z_2, \dots)$, has no eigenvalues. Which step of the finite-dimensional existence proof fails?*

Problem 7.10. *Verify that all matrices of a fixed linear transformation have the same determinant, and the same characteristic polynomial. Show the converse fails: exhibit two matrices with equal characteristic polynomials that are not similar.*

Problem 7.11. *Let $T \in \mathcal{L}(V)$ and let u, w be eigenvectors of T such that $u + w$ is also an eigenvector. Prove u and w belong to the same eigenvalue. Prove further that $v_1, \dots, v_m \in V$ are linearly independent if and only if some $T \in \mathcal{L}(V)$ has them as eigenvectors for distinct eigenvalues, and deduce that $e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ are linearly independent for distinct real λ_i.*

Problem 7.12. *A circulant C is defined by $C_{jk} = c_{(k-j) \bmod n}$. Show that the columns of the DFT matrix (Sheet 2, Problem 5) are eigenvectors of every circulant, and compute the eigenvalues. Deduce that the adjacency matrix of the cycle C_n has eigenvalues $2\cos(2\pi k/n)$ for $k = 0, \dots, n-1$.*

Problem 7.13. *Let A be the adjacency matrix of a simple graph G on n vertices with eigenvalues $\lambda_1, \dots, \lambda_n$. Show that $(A^k)_{uv}$ counts walks of length k from u to v, that $\sum_i \lambda_i^k$ counts closed walks of length k, and that G has $\frac{1}{6}\sum_i \lambda_i^3$ triangles. If G is d-regular, show $\lambda_1 = d$ and $|\lambda_i| \le d$ for all i.*

Problem 7.14. *Let $P \in M_n(\mathbb{R})$ have nonnegative entries. Show that 1 is an eigenvalue if every row sums to 1, and also if every column sums to 1. Show that every eigenvalue satisfies $|\lambda| \le 1$, and that P can have an eigenvalue of modulus 1 other than 1 itself.*

Problem 7.15. *Let $A, B \in M_n(\mathbb{C})$ with $AB = BA$. Prove that A and B have a common eigenvector. Does the conclusion survive over $\mathbb{R}$?*

Problem 7.16. *Give a real matrix with no real eigenvalue. Prove that every $A \in M_n(\mathbb{R})$ with n odd has one. Prove that every $A \in M_n(\mathbb{R})$ has an invariant subspace of dimension 1 or 2.*

Problem 7.17. *True or false, with proof or counterexample, for operators on a finite-dimensional space.*

1. *If v is an eigenvector of $\mathcal{A}^2$, then v is an eigenvector of $\mathcal{A}$.*
2. *If 0 is an eigenvalue of $\mathcal{A}^2$, then 0 is an eigenvalue of $\mathcal{A}$.*
3. *If $\mu^2 = \lambda$ and λ is an eigenvalue of $\mathcal{A}^2$, then μ or $-\mu$ is an eigenvalue of $\mathcal{A}$.*

Problem 7.18. *True or false, with proof or counterexample.*

1. *If $\mathcal{A} + \mathcal{B} = \mathcal{I}$, then $\mathcal{A}$ and $\mathcal{B}$ have the same eigenvectors.*
2. *If $\mathcal{A}\mathcal{B} = \mathcal{O}$, then $\mathcal{A}$ and $\mathcal{B}$ have the same eigenvectors.*
3. *$\mathcal{A}^2 = \mathcal{O}$ if and only if 0 is the only eigenvalue of $\mathcal{A}$.*
4. *$\mathcal{A} \neq \mathcal{O}$ is a zero divisor if and only if 0 is an eigenvalue of $\mathcal{A}$.*
5. *Every eigenvector of $\mathcal{A}$ lies in $\operatorname{Ker}\mathcal{A}$ or in $\operatorname{Im}\mathcal{A}$.*
6. *$\mathcal{A}^2 = \mathcal{A} \neq \mathcal{O}$ if and only if $\operatorname{Im}\mathcal{A}$ is an eigenspace of $\mathcal{A}$.*